

GULLFAX PLATFORM, Norway

WMO: 013755

Lat: 61.033N

Lon: 2.033E

Elev: 10

StdP: 101.20

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.8	0.2	-8.2	1.9	1.3	-7.0	2.1	1.9	31.2	4.1	29.0	4.6	13.5	330	1.171

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.3	18.2	15.1	16.9	14.7	15.8	14.1	16.1	17.6	15.2	16.4	14.3	15.3	8.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.2	10.8	16.3	14.8	10.5	15.8	13.9	9.9	14.9	45.3	17.1	42.3	16.1	39.8	15.3	19.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.3	24.5	22.2	-2.4	20.8	1.5	2.4	-3.5	22.6	-4.4	24.0	-5.2	25.4	-6.3	27.2
			-3.9	16.6	1.4	1.6	-4.9	17.7	-5.7	18.6	-6.5	19.5	-7.5	20.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	5.1	4.6	4.9	6.3	8.2	10.5	12.8	13.5	12.0	9.3	7.1	5.8	
	DBStd	3.64	1.85	2.18	1.98	1.86	1.95	1.75	2.05	1.78	1.80	1.85	1.99	2.25	
	HDD10.0	902	152	151	158	111	63	12	1	0	4	33	87	132	
	HDD18.3	3640	410	384	416	359	315	235	173	152	191	278	336	390	
	CDD10.0	304	0	0	0	1	6	27	87	107	63	12	1	0	
	CDD18.3	2	0	0	0	0	0	0	1	1	0	0	0	0	
	CDH23.3	1	0	0	0	0	0	0	1	0	0	0	0	0	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	11.4	14.1	13.3	12.2	10.7	9.8	9.5	8.9	9.2	10.9	12.3	12.8	13.2	
Precipitation	PrecAvg	1341	161	132	127	72	48	65	81	100	117	141	139	156	
	PrecMax	1652	252	257	237	123	83	132	153	162	190	216	220	240	
	PrecMin	954	76	58	46	26	22	26	40	66	48	58	43	65	
	PrecStd	165	48	53	46	23	16	24	22	25	34	41	39	42	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.2	9.2	9.2	12.2	16.1	18.8	21.8	20.2	17.8	13.2	11.2	10.2	
		MCWB	8.6	8.6	7.9	8.1	11.4	13.7	16.1	15.2	14.2	11.2	10.1	9.6	
	2%	DB	9.0	8.8	8.2	10.1	13.1	15.2	19.0	18.0	16.0	12.8	10.8	9.8	
		MCWB	8.5	8.2	7.3	7.6	9.7	12.5	15.6	15.5	13.7	11.1	9.8	9.3	
	5%	DB	8.2	8.1	8.0	9.1	11.2	13.2	16.8	16.9	15.0	12.1	10.1	9.1	
		MCWB	7.4	7.4	7.1	7.6	9.1	11.3	14.8	15.3	13.2	10.6	9.2	8.5	
	10%	DB	7.9	7.8	7.2	8.2	10.2	12.2	15.2	15.9	14.1	11.8	9.9	8.8	
		MCWB	7.1	6.9	6.2	7.1	8.8	10.7	13.9	14.4	12.8	10.4	8.9	8.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.0	9.1	8.6	9.2	12.2	14.6	17.3	17.5	15.2	12.6	11.0	10.0
			MCDB	9.1	9.2	8.7	10.0	15.4	17.3	20.2	18.5	16.5	12.9	11.1	10.1
2%		WB	8.3	8.1	8.0	8.5	10.7	12.7	16.1	16.4	14.2	11.9	10.2	9.2	
		MCDB	8.4	8.3	8.1	9.1	12.2	14.8	18.0	17.5	15.2	12.3	10.3	9.2	
5%		WB	7.6	7.5	7.2	8.0	9.9	12.0	15.2	15.4	13.8	11.3	9.6	8.6	
		MCDB	8.0	7.8	7.5	8.6	10.7	13.2	16.6	16.4	14.7	11.9	9.8	8.8	
10%		WB	6.9	6.7	6.6	7.5	9.2	11.4	14.2	14.6	13.2	10.7	8.9	7.9	
		MCDB	7.5	7.2	7.1	8.1	10.0	12.4	15.3	15.5	14.1	11.5	9.3	8.4	
Mean Daily Temperature Range	5% DB	MDBR	2.7	2.8	2.7	2.6	2.5	2.3	2.3	2.3	2.3	2.2	2.5	2.6	
		MCDBR	3.0	3.1	2.8	3.2	3.8	3.7	4.2	3.8	3.1	2.5	2.5	2.7	
		MCWBR	3.5	3.7	3.2	2.5	2.5	2.5	2.5	2.6	2.4	2.5	3.1	3.6	
	5% WB	MCDBR	2.8	3.1	2.5	2.6	3.0	3.3	3.5	3.2	2.6	2.2	2.5	2.7	
		MCWBR	3.7	3.9	3.3	2.6	2.4	2.5	2.5	2.7	2.5	2.9	3.5	3.7	
Clear-Sky Solar Irradiance	taub		0.334	0.355	0.368	0.382	0.370	0.363	0.381	0.372	0.365	0.347	0.321	0.385	
	taud		2.070	2.193	2.271	2.302	2.409	2.469	2.438	2.461	2.430	2.312	2.136	2.312	
	Ebn at Noon		350	599	740	809	853	867	838	816	747	609	366	222	
	Edh at Noon		45	72	91	106	103	99	100	90	79	63	40	29	
All-Sky Solar Radiation	RadAvg		0.19	0.69	1.84	3.59	5.02	5.15	4.66	3.67	2.22	1.04	0.30	0.12	
	RadStd		0.03	0.10	0.23	0.33	0.34	0.56	0.39	0.32	0.18	0.13	0.04	0.01	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	+0.75	N/A	N/A	N/A	N/A	-99	-142	N/A	N/A
Regional (24 neighbors)	N/A	+0.65	N/A	N/A	N/A	N/A	-83	-118	N/A	N/A

Nomenclature: See separate page